

New Møller Target Movement Control

12 January 2010

Target Frame Parameters

1. The targets are round foils with diameter **24.0mm** see Picture 3. Target frame scheme (PDF appended).
2. The distance between the foils centers of is: **1.5"**.

Linear Moving System

The stepper motor and the limit switches are connected to T-Channel of the motor controller VME44.
Basic Parameters:

1. Number of positions : 6
 - 2 limit positions: "Retracted Limit Switch" (no target on the beam, target is parked) and "Extended Limit Switch";
 - 4 positions (a target in the beam): "Target-1", "Target-2", "Target-3", "Target-4" (numbering from "Retracted Limit Switch" to the "Extended Limit Switch" see Picture 3).
 - Each of the 6 positions is determined by switches that are connected to the stepper motor controller and by the encoder (position transducer)- see the specs. Output signal from the encoder is a positive voltage potential in a range of 0-4V. The encoder signal can be readed from EPICS variable HAMOL_ENC1.
2. Range of motion from the "Retracted Limit Switch" to the "Extended Limit Switch" is: **356 mm**
3. Travel distance per one stepper motor revolution is: **~0.36 mm/revolution**
4. The number of steps per revolution is: **200**
5. In microstepping mode there is **10 microsteps/step**
6. The movement speed of the target frame is: **~ 8-10 mm/sec**

Linear Limit Switches (see Table 1):

1. All the limits switches are "Normal Open" (gives logical(TTL) "1" to the controller) when they are not pressed. When the limit switch is pressed (a target in position or in one of limit position) it is closed and the logical "0" appears on the input of the controller.
2. "Extended Limit Switch" is connected to the T-Channel (pin **23C** of backplane VME44 connector)
3. "Retracted Limit Switch" is connected to the T-Channel (pin **24A** of backplane VME44 connector)
4. All the targets switches are connected in parallel and look like one switch, which is connected to the "Home Limit Switch" of the T-Channel (pin **24C** of backplane VME44 connector). The software should count the target switch crossings to cross check with the encoder reading for the target position.

To enable the stepper motors operations the "T Auxiliary Output"(**22C**) should be set to logical "0"

Signals to FSD System

1. We suggest two modes of FSD operations:
 - a. **Normal mode** an FSD signal is generated if the target frame is not in a "Target" or "Retracted Switch Limit" position.
 - b. **Møller mode** The Møller FSD signal is not masked. This mode should be used only during Møller measurements. The target can be moved only with the beam turned off.

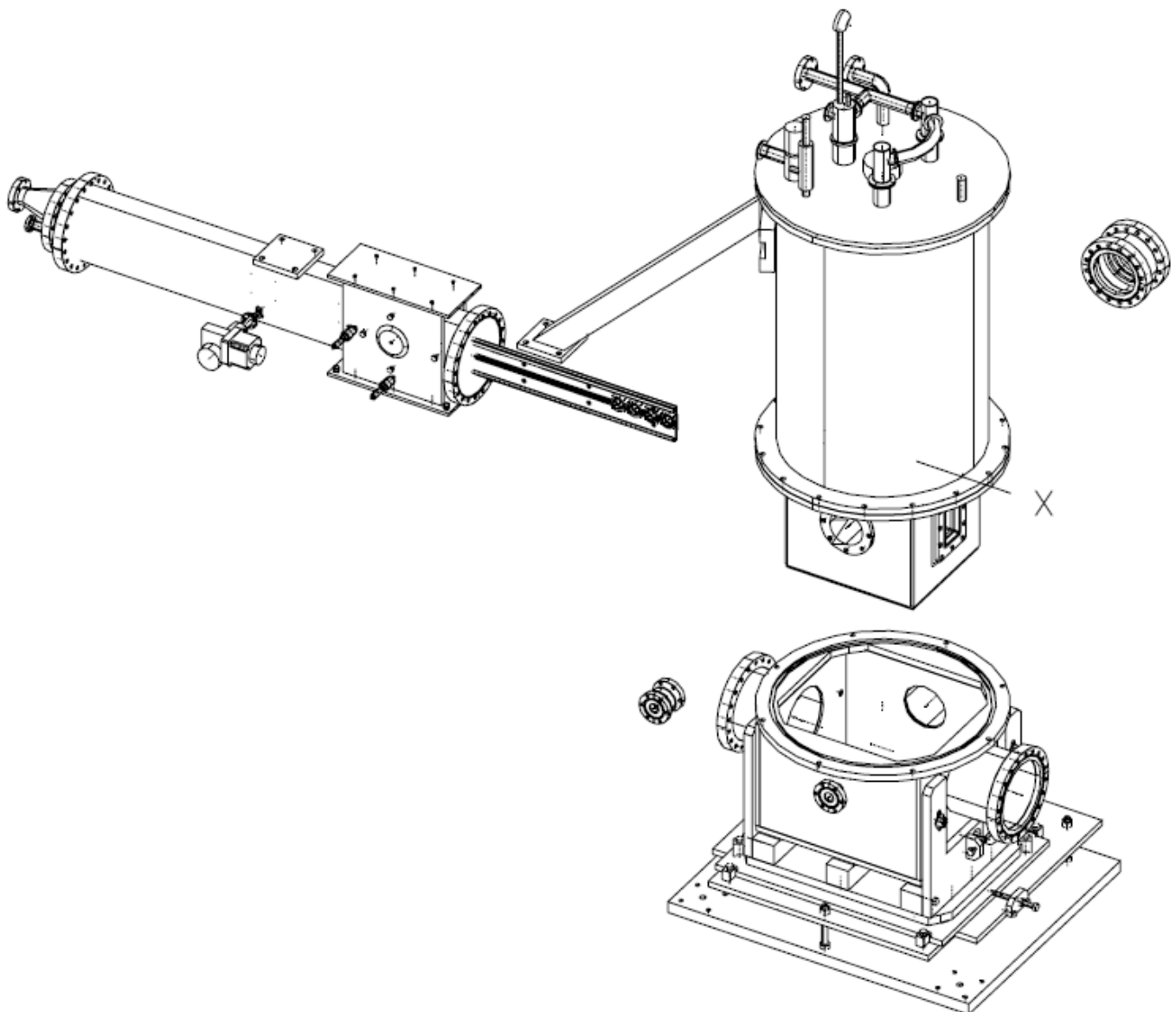
Software Requirements

1. Initialization: It has to be able to initialize system any time without iochla reboot
 - a. Move the linear stepper motor to the "Retracted Limit Switch" of target frame and set this as the HOME position.
2. Normal Operation Requirements:
 - a. Move to Target 1-4 : a routine that moves the requested target on the beam.
 - b. Move to Home Position ("Parking Limit Switch"): a routine that moves the target frame "Hole" on the beam.

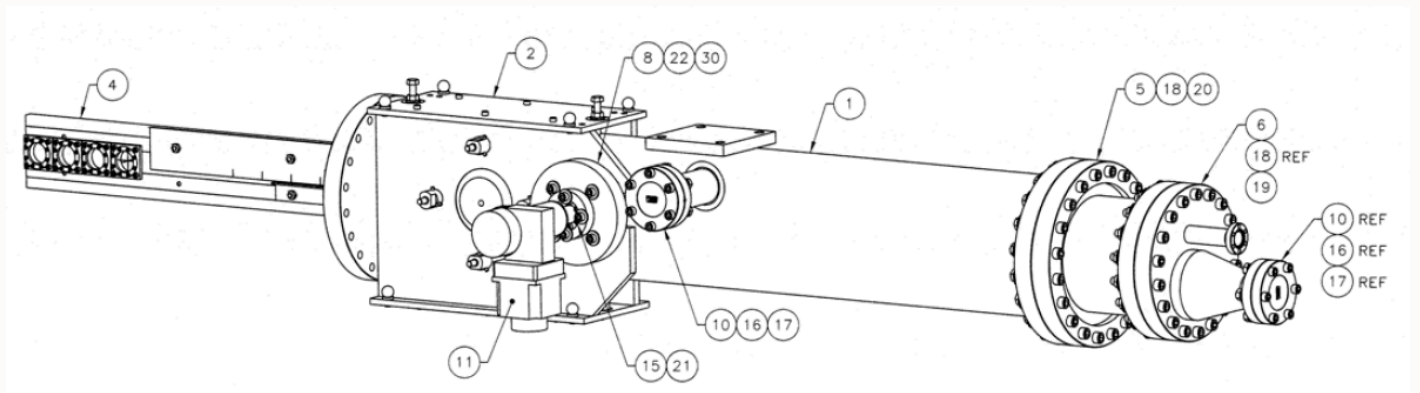
Software interlocks: there is an interlocks for movement of the target frame on or out of the beam.

1. Only MCC can move target after the Hall A request.
2. Off beam for any target movement.
3. Beam is available only if the target position is on the "Parking Limit Switch" or on any "Target-N".

Picture 1. View of Møller Polarimeter Target Assembly.



Picture 2. View of Target Loader Assembly.



Picture 3. View of Target Holder Assembly.

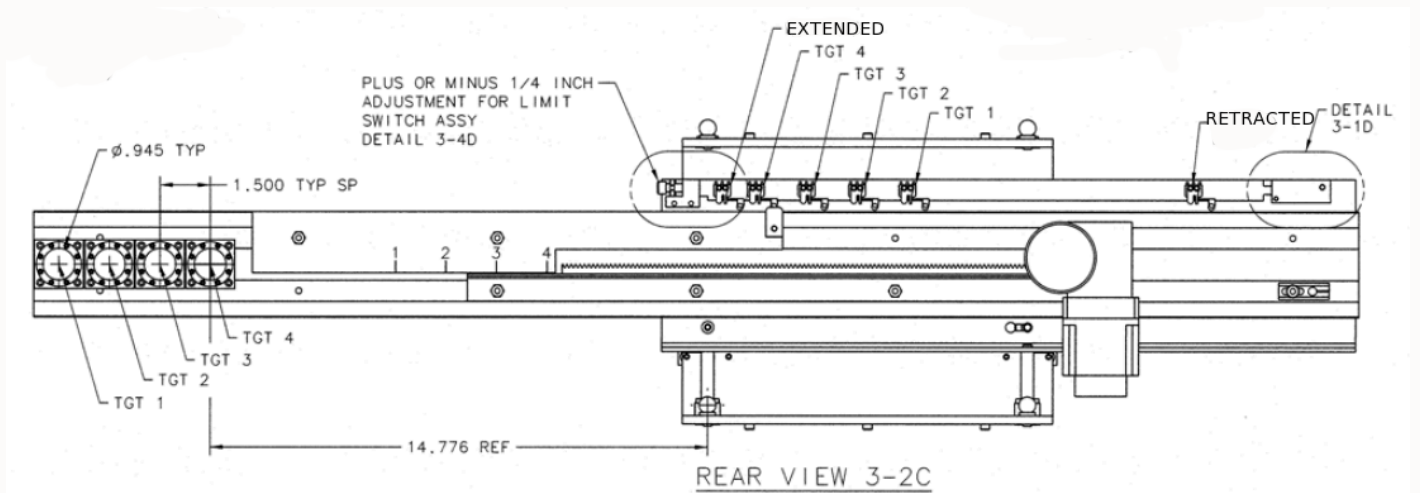
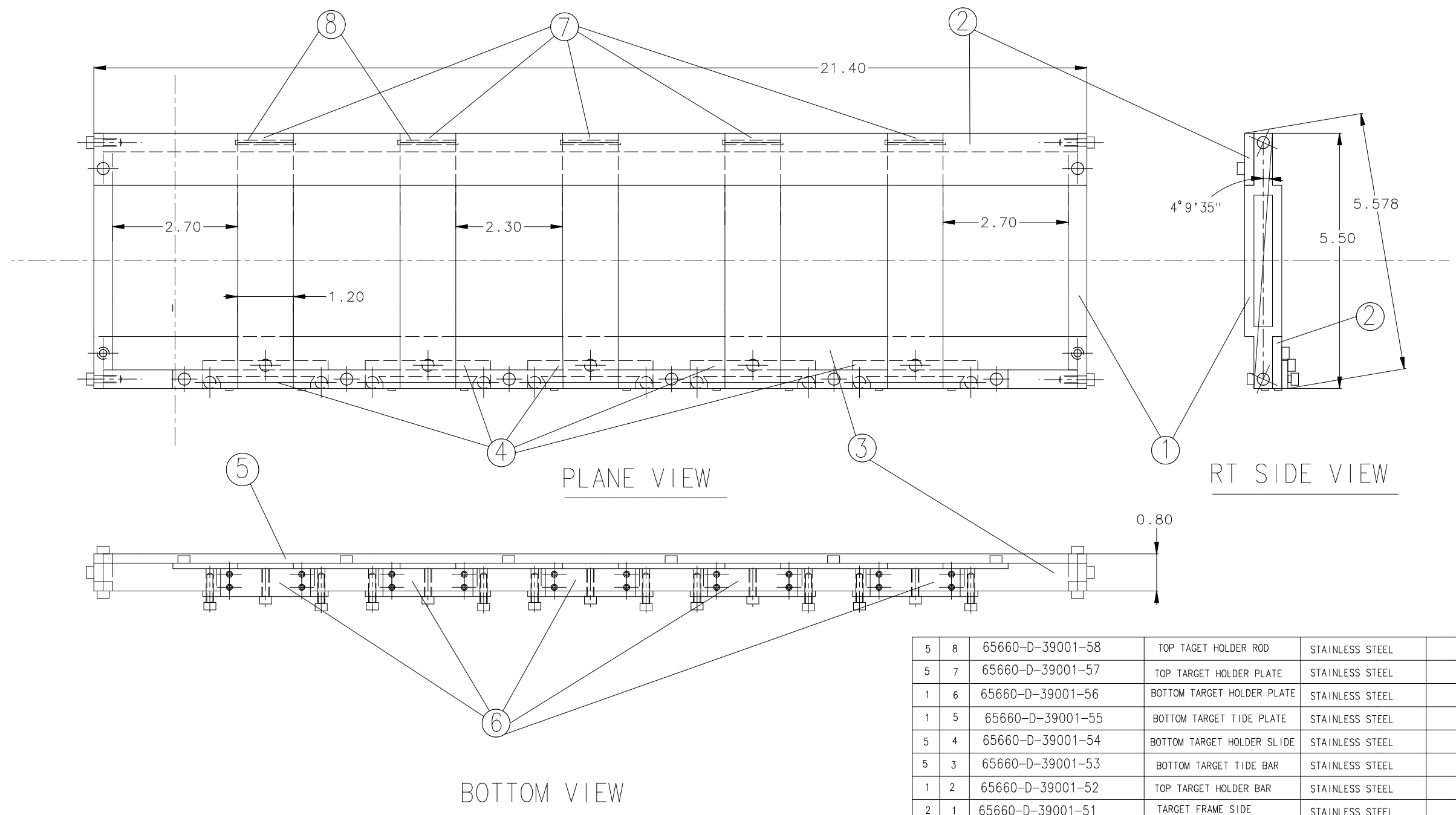


Table 1. Pinout OMS Chassis to Stepper Motor Controller VME44

Signal Name	P2 OMS Board	P5 Connector	Color	OMS Chassis
+5 Volts Channel T	21A	b	red	+5V to linear MD125(10pin)
T Step Output	21C	c	blue	step pulses to linear MD125(9pin)
T Direction Output	22A	d	yellow	direction to linear MD125(8pin)
T Auxiliary Output	22C	e	green	"0"-enable Linear Motor
Ground	23A	f	black	Common
T Extended Limit Switch	23C	g	orange	Linear Right Limit Switch
T Retracted Limit Switch	24A	h	brown	Linear Left Limit Switch
T Home Limit Switch	24C	j	white	Linear Target On Beam Switch



5	8	65660-D-39001-58	TOP TAGEY HOLDER ROD	STAINLESS STEEL	
5	7	65660-D-39001-57	TOP TARGET HOLDER PLATE	STAINLESS STEEL	
1	6	65660-D-39001-56	BOTTOM TARGET HOLDER PLATE	STAINLESS STEEL	
1	5	65660-D-39001-55	BOTTOM TARGET TIDE PLATE	STAINLESS STEEL	
5	4	65660-D-39001-54	BOTTOM TARGET HOLDER SLIDE	STAINLESS STEEL	
5	3	65660-D-39001-53	BOTTOM TARGET TIDE BAR	STAINLESS STEEL	
1	2	65660-D-39001-52	TOP TARGET HOLDER BAR	STAINLESS STEEL	
2	1	65660-D-39001-51	TARGET FRAME SIDE	STAINLESS STEEL	
QTY. REQD.	ITEM NO.	PART OR IDENTIFYING NO.	NOMENCLATURE OR DESCRIPTION	MATERIAL SPECIFICATION	NOTES
DIM & TOL OER ANSI Y14.5 UNLESS OTHERWISE SPECIFIED DIMENSION ARE IN INCHES TOLERANCES ARE: FRACTIONS DECIMAL ANGLES 1/16 .XX .015 0.50 .XXX .005			CADD I.D. NO.	The Thomas Jefferson National Accelerator Facility NEWPORT NEWS, VIRGINIA UNITED STATES DEPARTMENT OF ENERGY	
MATERIAL			APPROVALS	DATE	HALL A MOLLER POLARIMETER SCANNING TARGET ASSEMBLY TARGET FRAME
STAINLESS STEEL			DRAWN V.GORBENKO		
			CHECKED		
FINISH			APPROVED/ORIGINATOR		SIZE D
12.5 UNLESS OTHERWISE NOTED DEBURR & BREAK ALL SHARP EDGES			APPROVED		
DO NOT SCALE DRAWING					DWG. NO. 65660-D-39001-50 REV. SCALE 1:1 USED ON ASSY NO. 65660-D-39001-40 SHEET 1 OF 1